

ME597 IIOT Project Template

Directions: Please click "**File**" and select "**Save a copy in Drive.**" Then, share this with all of your teammates using their gmail addresses. Finally, please fill in your group number project title, and team members, below.

Group Number: 6

Project Title: Automated Coffee Maker Monitoring

Team members: R. Daniel Claud IV

Milestone 0: Project Charter (Complete by End of Week 4)

0.1 Initial Project Description

[Please provide a high level overview of the of project you hope to complete in this course. Please make sure to provide the following elements: The company you want to work with and the scope of the project]

The project is to develop a monitoring system for a coffee maker. This will include temperature and state monitoring to determine when the coffee maker is turned on, and if a fresh pot of coffee is ready for consumption. It will also include a time estimate of how long a pot of coffee has been sitting ready for someone to drink it.

0.2 Physical Asset

[Please provide a detailed description of the physical asset that you are going to monitor. Please provide pictures, figures, schematics, and any other relevant information you have at this time.]



The physical asset to be monitored for this project is a simple Gormia drip pot coffee maker. It contains a water tank at the back that is filled through the top. It also features a timer that can be set to turn the maker on at a preprogrammed time. Finally it also includes a heating element at the bottom to keep the coffee warmer for a longer period of time.

0.3 Asset Role

[What role does it play in the big picture of the company?]

This project is a self-led project of a home device. The role that it plays in my own life is that it keeps me caffinated and gives me a boost of energy in the morning.

0.4 Company Benefits

[Please provide a brief expalantion of how your project will provide a benefit to the company in question?]

This project will allow me to gain more insight into how long it takes to make coffee, how fresh a pot of coffee is, and will be able to give me monitoring insights into when the coffee maker is running. This will help me a lot since I am prone to walking away from my coffee machine or forgetting that I have set a timer for it.

0.5 Questions or Concerns?

[What questions or concerns do you have about this project that you would like eedback on?]

I would like feedback on other ways to monitor the device other than the sensors provided for lab.

Milestone 1: Introduction and Planning (Complete by End of Week 7)

1.1 Project initiation

[Please describe the process for being assigned to this project and your role in the decision-making process. Please provide details of any initial contact and communication with the company. What was the conversation about?]

This project was started after two failed attempts at projects with other companies

- First Project: Particle Monitoring System for Nothrop Grumman
 - Unsuccessful due to cybersecurity policies related to working in the defense industry
 - The timeline for this class was not long enough to account for sufficient cybersecurity review and approval by the company
- Second Project: Mobile Test Station for Dormakaba
 - Unsuccessful due to a lack of communication with assigned partner
 - Was unable to contact appropriate stakeholders

After these two failed attempts it was decided that I should move forward with doing a project on a home device.

1.2 Project description

[Describe what the project consists of in terms of (a) the work that needs to be done, (b) the technology or devices that will be used, and (c) what problem will be solved for the company. What ethical or privacy considerations have you pondered?]

Work that needs to be done

1. Develop appropriate mounts for the sensors onto the coffee maker
2. Write Python scripts to interface with the sensors and collect data
3. Develop Machine Learning models to detect when coffee is brewed

Technology and devices used

1. DS18B20 Thermocouple to monitor how hot the water tank gets while brewing coffee
2. ADXL345 Accelerometer to monitor the pumping of water from the water tank to the pot above the grounds

Problem that will be solved

Identifying when my coffee is done in case I walk away.

1.3 Target audience

[Who are or will be the users of the project innovation? What are their needs? What are their backgrounds and expertise? How have their input been considered in the project planning and implementation stages?]

I am the primary target audience. Outside of this, the target audience includes anyone who loves coffee and is super busy and prone to walking away while coffee is being brewed.

The customer needs include a system to notify and monitor the status of a fresh batch of coffee.

My background and expertise for this project are that I am a busy professional engineer and part time graduate student. My own input has been considered for shaping the entirety of the project.

1.4 Roles and responsibilities

[Please describe your team and members' expertise. How was your team formed? What are team members' roles and responsibilities? What are the involved parties in this project, and what are their roles? For example, will your course instructor and TA have a role? Who are the other participating stakeholders (e.g., managers, operators), and what would be their roles?]

Expertise

Microelectronics Semiconductor Engineer working at Northrop Grumman

Day-to-day tasks as a full-stack software engineer

- Developing frontend web applications using TypeScript and React
- Developing backend APIs in Python using Flask or FastAPI frameworks
- Networking with industrial automation sensors and devices catered for the Semiconductor Industry

Mechanical Hardware Engineer

- Performed structural analysis on components for use in undersea environments
- Developed CAD models for electronic enclosures

Responsibilities

I will be handling the entirety of the project. I will be responsible for:

- Acquiring appropriate mounts for the sensors
 - Designing in CAD and 3D printing if needed
 - Writing a Python script to read the data using MTConnect
 - Developing a relational database schema using MySQL
 - Developing a user interface either in Grafana or a full web application using TypeScript/React
 - Developing machine learning models to notify when coffee is brewed I will need support from the TA Yuseop Sim as a resource and mentor for this project
 - Acting in an advisory or consulting role
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1.5 Communication channels

[What is the process that you and the stakeholders have defined for keeping the communication going? For example, who will be in contact with the company? What is their role? How often will you communicate? How will you communicate?]

I will communicate on an as-needed basis with the professor and TA using teams or outlook emails.

1.6 Resources

[What resources (hardware, software, server, and so on) do you plan to use? How will you acquire those resources? (e.g., company, funding project, school, or others: open source, free trial, etc.)

Frontend Development

- Typescript
- React Framework
- Vite Bundler

Relational Database

- Schema developed in MySQL

Machine Learning Models

- Models developed using TensorFlow
- Will utilize Jupyter Notebooks for writing algorithms

Hardware

- ADXL345 Triple-Axis Accelerometer
 - DS18B20 Thermocouple
 - Raspberry PI
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1.7 Tentative tasks and proposed timeline

[What specific activities do you as a team need to accomplish, and who will be the person in charge? What is the timeline that was set for this project? What was your role in defining the scope of the work and the timeline? What are the internal milestones you have considered?]

Sensor Deployment – Week 11

- Develop mounts to affix sensors to the coffee machine
- Develop a database schema to parse data into

Frontend Development – Week 13

- Develop a front-end application to track sensor readings in real-time

Machine learning modelling – Week 16

- Develop a machine learning model

Automated notifications – Reach Goal

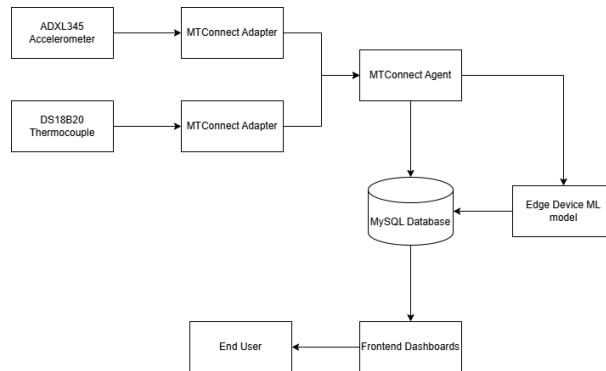
- Assign email credentials to the monitoring system to send an email once coffee is brewed
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Milestone 2: Project Implementation and Evaluation (Complete by end of Week 11)

2.1 Technical implementation details

[From a technical perspective, please describe a preliminary plan to implement your solution to address the problem to be solved. Please provide any figures, pictures, or diagrams of your proposed solution. You can also include pictures of the devices to be used and the actual machinery.]

The preliminary plan to implement the solution is to attach an accelerometer to the outside of the coffee maker to monitor the water pump inside the coffee maker. In addition, I will use a thermocouple to monitor the temperature of the coffee pot. Combining these two raw sensor outputs I will develop a machine learning model to determine the state of the coffee maker and a frontend dashboard to communicate results.



2.2 Evaluation and testing

[What validation or verification procedures have the team established to ensure that the solution works properly? What are the results of initial testing? Has testing been performed with the target users? What was the outcome of the test? What are unforeseen factors that need to be addressed?]

I have developed a validation procedure where the output of the ML model is monitored while the coffee machine is turned on, as well as when it is turned off to ensure that the model correctly predicts the state of the coffee machine. I am not verifying the reading of the thermocouple and am simply assuming that its readings are correct. The preliminary tests of the thermocouple have been successful and it is possible to train a model on acceleration data.

2.3 Activities and progress

[What major activities or tasks has the team accomplished? What are activities that remain to be completed? How is the overall progress?]

I have collected sensor data using the ADXL345 and trained preliminary models on it. I still need to deploy the thermocouple and build a dashboard for it.

2.4 Expected outcome

[Please describe the current state of the operation and the tentative benefit the company will gain after implementing your proposed solution. What is the primary concern? What do you envision are the challenges for this implementation? What are the challenges for permanently adopting this proposed solution?]

The current state is that not much permanent progress has been made, since the hardware is needed for the lab activities. The tentative benefit I will get from the proposed solution is that I will be able to better monitor the state of my coffee machine. The primary concern is that coffee gets cold or goes bad or is forgotten about while I am busy. I envision that the challenges for the implementation is wiring everything up in a consistent way and storing the data reliably. The challenges for permanently adopting the solution are that the wiring setup is quite amateurish and can easily come disconnected.

2.5 Division of labor

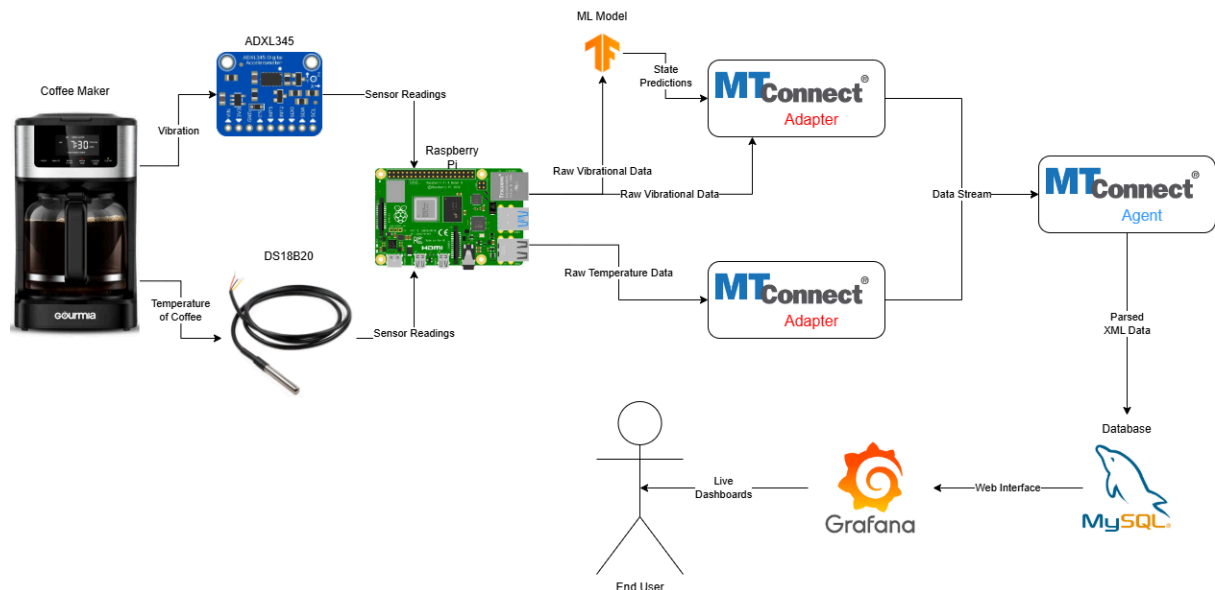
[How has the team evolved from the beginning of the semester in terms of cohesion and trust? How have roles shifted or refined? How is the interaction with the contact person from the company? How frequent is the communication and ongoing feedback?]

Not applicable since this is a self-led project where I am the only team member and I do not communicate with a company.

Milestone 3: Project Solution, Recommendations and Structure Technical Report (Complete by end of Week 15)

3.1 Project solution

[Show and explain your solution. Interpret the output and show how the proposed solution addressed the problem/project. Identify limitations along with ranges of operation.]



The final solution consists of an ADXL345 Accelerometer and a DS18B20 Thermocouple attached to the coffee maker in strategic places: the ADXL345 firmly secured to the side of the machine and the DS18B20 secured to the side of the pot. These two sensors send raw information to the raspberry pi. The raspberry pi takes the raw acceleration data and sends it to a machine learning model trained to determine one of three states:

1. OFF, UNAVAILABLE - The coffee maker is idle, not brewing any coffee
2. ON, BREWING - The coffee maker is in normal operation but won't be done for some time
3. ON, FINISHING - The coffee maker is finishing up and the coffee will be ready to drink soon

The temperature sensor is included to judge how fresh or warm the coffee is before the user drinks it.

The raw acceleration and temperature data as well as the state predictions of the model are sent through MTConnect to a MySQL database. This can then be viewed on a Grafana dashboard which can give the user insight into the current state of the coffee pot and how fresh the coffee is.

The limitations of this project are as follows.

1. The thermocouple is connected directly to the coffee pot. This can make the coffee difficult to pour and means that the pot must be in close proximity to the machine
2. The electronics are exposed. This means that if a spill were to occur, the electronics might get damaged.
3. The ML model is biased towards the ON, BREWING condition rather than the ON, FINISHING condition. This means that the incorrect state can be displayed at times.

Overall, the proposed solution does in fact give insight into the status of a coffee maker with good accuracy and allows the user to move away and do other tasks without needing to watch the pot. This allows the coffee to be drunk at optimal freshness and improves the user experience.

3.2 Anticipated benefits and adoption

[How do you envision the major benefit the company will receive after your implementation of this project? Do you expect your project solution to be permanently adopted by the company? What would be the major challenges for permanently adopting this solution?]

As this is a home-project, I will be the primary beneficiary of this project. The main benefit I will receive after my implementation is better tracking of my coffee machine. This allows me to go and do other things in my apartment such as do the dishes, go downstairs to check the mail, and other tasks while waiting for the coffee to brew. I expect that I will

continue to use this solution, since I have found it to be quite convenient. However, the major challenge I foresee with using the solution is the exposed electronics to water, since it poses a risk of electrical hazard. The other challenge I foresee is the connected thermocouple to the pot which makes the range of motion a bit difficult to use.

3.3 Human factors and Impact

[How will your proposed solution impact the worker or operator directly interacting with your project solution in terms of benefits and challenges? How will the nature of the job of other workers or users change as a result of your project solution?]

The proposed solution impacts the operator (myself) by improving the insights into the coffee maker's operational state and current temperature. However, it physically restricts the movement of the coffee pot making it difficult to pour or fill with water. This requires a secondary container to fill the internal tank with water. This will change my job of checking on the coffee pot as I won't have to monitor it very often and can just monitor the pot from a second room.

3.4 Personal learnings and takeaways

[Please respond to this question individually by starting the response with your initials. What were your expectations about your project solution, and how did those shift until this point? How has this experience impacted your views about the manufacturing industry? What key learnings, takeaways, or personal growth have you gained after this project experience? What was it like to work with community or company members? What perspectives have shifted, or have you become aware as you interacted with the company? How has your relationship with the company evolved over time?]

Initials: RDCIV

My expectations of the project solution were that the machine learning would be very difficult to accomplish and that putting all of the pieces together would be exceptionally difficult. These started to shift as I completed more labs related to machine learning, specifically the labs with the AFF and I started to actually understand how machine learning works. After this project experience, I have a new appreciation for the tiny Raspberry Pi, since I now know just how capable it is. Additionally, I am no longer scared by the concept of machine learning since I understand it now. Since this was an at-home project, I did not need to work with community or company members. I did not gain any insights into these perspectives.

3.5 Next steps and recommendations

[What are other recommendations you would like to offer that would benefit the company? What are the missing steps to make your project implementation fully functional?]

The recommendations I would like to offer that would benefit the company include using a bluetooth thermocouple or moving the thermocouple to a different location so as to not impede the movement of the coffee pot. I would also recommend including additional sensors such as a hall sensor or ultrasonic sensor to detect the presence of the coffee pot. Finally, I would recommend obtaining more training and validation data to improve the model's performance for indicating what state the coffee pot is in.

Structure Technical Report (Complete by end of Week 15)

Prepare this document to be used by the company for training purposes. Please provide as much detail as possible so someone unfamiliar with the project solution will be able to follow a standard procedure to implement and use your proposed solution.

1. Describe the components of the project solution. Provide pictures of the parts or components.
2. Provide a step-by-step procedure of how the solution should be installed and configured. Provide a video or picture of the ensemble.
3. Describe the functionality of the project solution and provide test cases and samples of outputs. If available, provide a video of the equipment working properly.
4. Provide a sample dataset for calibration and testing and provide the corresponding outputs.
5. Provide samples of interpretation of the outputs.

Place your code and upload it into project folder

*** Here's an example code snippet for citing a figure or video from an internal directory**

- Upload image and video from google drive

No description has been provided for this image

- Add figure name and description
- Add video as below and add description

```
In [ ]: from IPython.display import HTML

video_url = 'https://drive.google.com/file/d/14ot6CDCiimEHOCp9xf4kFkfjHiFuQ4Xl/preview'

html_code = f'<iframe width="720" height="480" src="{video_url}" frameborder="0" allowfullscreen></iframe>'
HTML(html_code)
```

Out[]:

```
In [ ]: !pip install pysoundfile
        !pip install bitstring
```

```

In [ ]: import requests
        from IPython.display import Audio
        import wave

        def get_download_url(gdrive_url):
            file_id = gdrive_url.split("/file/d/")[1].split("/")[0]
            download_url = f"https://drive.google.com/uc?id={file_id}"

            return download_url

        def play_wav_from_url(url):
            response = requests.get(url)
            with open("audio.wav", "wb") as f:
                f.write(response.content)

            with wave.open("audio.wav", 'rb') as wav_file:
                params = wav_file.getparams()
                audio = wav_file.readframes(params.nframes)
                audio_length = params.nframes / float(params.framerate)
                print(f"length: {audio_length} sec")
                return Audio(audio, rate=params.framerate)

        # Google Drive URL
        gdrive_url = "https://drive.google.com/file/d/1EbvW10fIt0KEJt2h_Han1g_YkRtE85Jr/preview"
        download_url = get_download_url(gdrive_url)
        play_wav_from_url(download_url)

```

length: 10.2 sec

Out[]:

